MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5636

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pentode, subminiature, dual control

Outline - - - - - - - - 3-1
Base - - - - - - - - E8-10
Envelope - - - - - - - - T3
Cathode - - - - - - - - Coated unipotential
Base connections:

Pin No. - - -	1	2	3	4	5	6	7	8
Element - - -	g1	k	h	g3	a	h	g2	k

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ec3	Ehk	Rk	Rg1	Ik	Ic2	Pp	Pg2	TE	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	v	Ohms	Meg	mAdc	mAdc	W	W	°C	ft
Maximum:	6.6	165	0	155	30	200	---	1.1	16.0	7.0	0.55	0.45	220	Note 9
Minimum:	6.0	---	-55	---	---	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	100	0	100	0	0	150	---	---	---	---	---	---	---

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GENERAL:

Qualification - Required

Reliable tube

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits		Unit
							Min	Max	
	Quality conformance inspection, part 1								
1256	Electrode current (1) (anode)			0.65	II	Ib	3.7	6.9	mAdc
1256	Electrode current (2) (anode)		Ec1 = -7.5 Vdc; Rk = 0	0.65	II	Ib	---	100	µAdc
1266	Total grid current	1	Rg1 = 1.0 Meg	0.65	II	Ic1	0	-0.3	µAdc
1301	Heater current			0.65	II	If	140	160	mA
1306	Transconductance (1)	1		0.65	II	Sm	2,700	4,000	µmhos
1336	Heater-cathode leakage			0.65	II	Ihk	---	5.0	µAdc
1201	Short and discontinuity detection			0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes			2.5	I	R	100	---	Meg
1031	High-frequency vibration		Rp = 10,000 ohms	2.5	I	Ep	---	60	mVac
1246	Audio frequency noise		Esig = 70 mVac; Ecc2 = 19 Vdc; Rp = 0.2 Meg; Rg1 = 0.1 Meg; Rg2 = 1,000 ohms	2.5	I	EB	---	17	vu
1256	Electrode current (3) (anode)	3	Ec3 = -8.0 Vdc	2.5	I	Ib	---	100	µAdc
1256	Electrode current (screen grid)			2.5	I	Ic2	2.8	5.4	mAdc
1256	Electrode current (suppressor grid)	3	Rg1 = 0; Rg3 = 0.1 Meg	2.5	I	Ic3	---	±0.1	µAdc
1266	Grid emission	4	Ef = 7.5 V; Ec1 = -7.5 Vdc; Rg1 = 1.0 Meg; Rk = 0	2.5	I	Ic1	0	-0.5	µAdc
1306	Transconductance (2)		Ef = 5.7 V	2.5	I	ΔSm / Ef	---	15	%
1306	Transconductance (3)	3	Ec3 = -1.0 Vdc	2.5	I	Sg3-p	500	1,800	µmhos
1331	Direct-interelectrode capacitance		Shield No. 318	6.5	Code F	Cg1p	---	0.020	pF
						Cg3p	---	1.10	pF
						Cg1g3	---	0.15	pF
						Cg1-all	3.5	4.5	pF
						Cg3-all	3.5	4.5	pF
						Cp-all	2.9	3.9	pF
1116	Lead fatigue			2.5	Code F	---	---	---	---
2126	Envelope strain			4.0	I	---	---	---	---
1041	Shock	5	450 G; Ehk = 100 Vdc	---	---	---	---	---	---
1031	Vibration-fatigue			6.5	Note 2	---	---	---	---

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 2 -Continued								
---	Shock and vibration-fatigue-test end points:								
1031	High-frequency vibration		Rp = 10,000 ohms	---	---	Ep	---	200	mVac
1336	Heater-cathode leakage			---	---	Ihk	---	20	µAdc
1306	Change in transconductance (1) of individual tubes			---	---	ΔSmt	---	20	%
1105	Permanence of marking			---	---	---	---	---	---

Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 3						
1506	Heater-cycling life		Ef = 7.0 V; 1 min "on", 4 min "off"; Ehk = 140 Vac	---	---	---	---
1516	Stability life		Ehk = 200 Vdc; Rg1 = 1.0 Meg; TA = room	---	---	---	---
---	Stability life-test end point:						
1306	Change in transconductance (1) of individual tubes			ΔSmt	---	15	%
1501	Intermittent life	6-8	Ehk = 200 Vdc; Rg1 = 1.0 Meg; TA = room	---	---	---	---
---	Intermittent life-test end point (500 hours):						
1211	Insulation of electrodes			R	50	---	Meg
---	Intermittent life-test end point (1,000 hours):						
1211	Insulation of electrodes			R	25	---	Meg
1501	Intermittent life (high temperature)	8	Ehk = 200 Vdc; Rg1 = 1.0 Meg; TA = room; TE = 220°C(min)	---	---	---	---
---	Intermittent life (high temperature)-test end points (500 hours):	10					
1266	Total grid current		Rg1 = 1.0 Meg	Ic1	0	-0.8	µAdc
1301	Heater current			If	138	164	mA
1256	Electrode current (suppressor grid)	3	Rg1 = 0; Rg3 = 0.1 Meg	Ic3	---	±0.1	µAdc
1306	Change in transconductance (1) of individual tubes			ΔSmt	---	20	%
1306	Transconductance (2)		Ef = 5.7 V	ΔSmEf	---	15	%
1336	Heater-cathode leakage			Ihk	---	10	µAdc
1211	Insulation of electrodes			R	50	---	Meg
1306	Transconductance (1) average change			AvgΔSmt	---	15	%

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Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 3 - Continued						
---	Intermittent life (high-temperature)-test end points (1,000 hours):	10					
1266	Total grid current		Rg1 = 1.0 Meg	Ic1	0	-0.8	uAdc
1301	Heater current			If	138	164	mA
1256	Electrode current (suppressor grid)			Ic3	---	±0.1	uAdc
1306	Change in transconductance (1) of individual tubes			ΔSm_t	---	25	%
1306	Transconductance (2)		Ef = 5.7 V	ΔSm_{Ef}	---	20	%
1336	Heater-cathode leakage			Ihk	---	10	uAdc
1211	Insulation of electrodes			R	25	---	Meg

NOTES:

1. This test shall be performed at the conclusion of the holding period.
2. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.
3. The reference point for Ec3 shall be the negative side of the cathode resistor.
4. Prior to this test, the tubes under test (TUT) shall be preheated for 5 minutes at the conditions specified below. The 3-minute test shall not be permitted. Test at specified conditions within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid emission test.

Ef	Ec1	Ec2	Ec3	Eb	Rk	Rg1
V	Vdc	Vdc	Vdc	Vdc	Ohms	Meg
7.5	0	100	0	100	150	1.0

5. A grid resistor of 0.1 megohm shall be added except when a thyratron-type short indicator is used.
6. The life-test sample shall consist of 20 tubes per lot and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample due to failure of more than 1 tube, a second sample of 40 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples. The total failures from the combined first and second samples shall not exceed three.
7. Eligibility for early release shall be as follows: Early release may be made at 500 hours providing no lot failure due to this life test or the regular 1,000-hour intermittent life test has occurred in the preceding three consecutive lots and this lot under consideration is acceptable at 500 hours.
8. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurements (method 1226), will be satisfied if a TUT having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.
9. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
10. The allowable defectives per characteristic shall not be greater than 1 for the first sample and shall not be greater than 3 for combined samples. Total defectives shall be 3 and 6, respectively, for 500 hours, and 4 and 8, respectively, for 1,000 hours.
11. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Air Force - 99
DLA - ES

User activities:
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DLA - ES

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